Stat 220 Midterm, A

Wait times at an urgent care clinic

A walk-in clinic installed a self check-in kiosk partway through the year. Management looked at average wait times before and after, concluded the kiosk made things worse, and is about to remove it. They have asked you to check.

Part 1. Multiple choice (40 points, 2 each)

Circle one answer for each question.

1. A clinic compares wait times on two days and gets $p = 0.21$. What has it shown?

   (A) That a difference this size is unsurprising if nothing is going on.

   (B) That the two days have the same average wait.

   (C) That the sample was too small for the test to work.

   (D) That the difference is real, but it is not large enough to matter.

2. Which of these does a 95% confidence interval for a difference in means tell you?

   (A) The range of differences the data leaves as plausible.

   (B) The range that contains 95% of individual patients.

   (C) The probability that the true difference is zero.

   (D) The range of values the sample difference would take 95% of the time.

3. A study with 40,000 patients finds a 0.3 minute difference in wait time with $p < 0.001$. What should you conclude?

   (A) The difference is real but far too small to matter operationally.

   (B) The test must be wrong, since such a small difference cannot be significant.

   (C) The effect is real and worth acting on, since the p-value is tiny.

   (D) The sample is too large for the p-value to be meaningful.

4. Which power statement is correct?

   (A) Power does not depend on the size of the effect you are looking for.

   (B) Power is the chance that the null hypothesis is true.

   (C) Power rises as the significance cutoff is made stricter.

   (D) Power is the chance of seeing a real effect when one is there.

5. A manager tests 20 unrelated operational changes at the 0.05 cutoff and finds one significant. What is the most likely explanation?

(A) On 20 useless tests, one significant result is expected by chance.

(B) The other 19 tests were underpowered.

(C) The data must have been collected incorrectly for the other 19.

(D) That change is the only one that worked.

6. The outcome is whether a patient leaves before being seen, yes or no. Which family fits?

(A) Linear regression, since the outcome is coded 0 and 1.

(B) Logistic regression, which returns a probability between 0 and 1.

(C) Poisson regression, since the outcome is a count of patients.

(D) A random forest, since the outcome is not continuous.

7. Which is the strongest reason to prefer a regression over a random forest when the question is whether a new kiosk changed wait times?

(A) A forest cannot handle categorical predictors like severity.

(B) A regression reports a coefficient with an interval, which is the answer.

(C) A forest will always predict less accurately than a regression does.

(D) A forest requires far more data than a regression does.

8. What does AIC do that R^2 does not?

(A) It measures how well the model predicts a new case.

(B) It works for any model that makes a prediction.

(C) It charges the model for each predictor it uses.

(D) It tests whether the coefficients are different from zero.

9. A regression of wait time on number of patients ahead gives a slope of 4.3. What does that mean?

(A) The model's predictions are off by about 4.3 minutes on a typical visit.

(B) Wait time is about 4.3 times the number of patients ahead.

(C) About 4.3% of the variation in wait time is explained by patients ahead.

(D) Each extra patient ahead is associated with about 4.3 more minutes of wait.

10. In `minutes ~ patients_ahead + staff_on_shift`, what does the coefficient on staffing mean?

(A) The effect of staffing, ignoring how busy the clinic is.

(B) The total effect of staffing, including its effect through patient volume.

(C) The effect of staffing among visits with the same number of patients ahead.

(D) The correlation between staffing and wait time.

11. A categorical predictor with four levels appears in the output with three coefficients. Why?

(A) One level was dropped for being statistically insignificant.

(B) Three is the maximum number of levels a model can estimate.

(C) The fourth coefficient is always zero and is omitted.

(D) One level is the baseline, folded into the intercept.

12. A coefficient on a treatment flips from positive to negative when a second predictor is added. What is the most reasonable conclusion?

(A) One of the two models is misspecified and should be discarded.

(B) The sign flip means the data contains an error.

(C) The flip is noise, since coefficients move around between models.

(D) Both are correct, and they answer different questions.

13. A residual plot shows a clear U shape. What does that say?

(A) There are outliers that should be removed before refitting.

(B) The residuals are not normal, so the outcome should be transformed.

(C) The relationship bends, so a straight line is the wrong shape.

(D) The predictors are too highly correlated with each other.

14. A clinic wants to tell one arriving patient how long they should expect to wait. Which interval do they need?

(A) The confidence interval for the fitted mean.

(B) Either one, since both come from the same model.

(C) The prediction interval for a new case.

(D) The confidence interval, because it is narrower and more precise.

15. As you collect far more data, what happens to the prediction interval for one new case?

(A) It narrows a little, then levels off near the spread of cases.

(B) It narrows toward zero width.

(C) It stays exactly the same width.

(D) It widens, because more unusual cases turn up in a bigger sample.

16. What makes error measured on the rows a model was fitted on too optimistic?

    (A) Those rows usually contain fewer recording mistakes than later ones.

    (B) The coefficients were chosen to make those rows' residuals small.

    (C) The sample is always smaller than the population.

    (D) The software computes it before the fit has converged.

17. Which is a fair description of what 5-fold cross-validation produces?

    (A) A guarantee of how the model will do next month.

    (B) An error estimate averaged over five held-out pieces of the data.

    (C) A p-value for whether the model beats a baseline.

    (D) A larger training set, which makes the model more accurate.

18. A model includes `kiosk`, `severity` and their interaction. What does the coefficient on `kiosk` by itself now mean?

    (A) The kiosk effect averaged over the three severity levels.

    (B) The kiosk effect among urgent visits only.

    (C) The kiosk effect with severity held at its average value.

    (D) The kiosk effect for visits at the baseline severity level.

19. Two predictors are correlated at 0.97. What should you expect?

    (A) Each will have a tiny p-value, since both predict the outcome well enough.

    (B) They explain little more than either alone, and both coefficients wobble.

    (C) The model will refuse to fit.

    (D) Dropping either one will sharply reduce predictive accuracy.

20. A model predicts wait times well. Management asks whether adding a third nurse would shorten waits. What is the right answer?

    (A) Yes, if the staffing coefficient is negative and significant.

    (B) Yes, since the model predicts accurately.

    (C) The data is observational, so it is a comparison, not an intervention.

    (D) No, because the model's R^2 is not high enough to support that.

Part 2. The analysis (60 points)

`clinic_visits.csv`: 900 completed visits.

Column	What it is
`minutes`	how long the patient was in the clinic, the outcome
`patients_ahead`	patients waiting when this one arrived
`waiting_room_count`	a second count taken by the front desk
`staff_on_shift`	clinicians working, 2 to 4
`severity`	minor, moderate, or urgent
`kiosk`	1 if the visit used the self check-in kiosk
`room_number, front_desk_rating, month`	also recorded

Load it with `pd.read_csv("https://drbob-richardson.github.io/stat220/F2026/data/clinic_visits.csv")`. Put each piece of work in the space provided.

1. **Which kind of model, and why** (6 points)

   Before you fit anything, decide what kind of model this question calls for. Name the family, say why it suits this outcome, and say what you need from it that a random forest could not give you.

2. **Look at the data first** (8 points)

   Make two plots that bear on the clinic's question. Then, in two or three sentences, say what you see that a later step will have to account for.

3. **The comparison management already made** (10 points)

   Management compared average wait times with and without the kiosk and decided the kiosk made things worse. Run that comparison yourself with a formal test. Report the results of the test in the appropriate context.

4. **Put other predictors in the model** (12 points)

   Other factor may be influencing the relationship between kiosk and wait times. Test the relationship between kiosk and wait times when considering other variables. Report your results appropriately.

5. **The category in the model** (6 points)

   Severity has three levels. Report its coefficients, say which level is the baseline and how you can tell, and write the sentence that interprets one of the others.

6. **Does the kiosk help more when the clinic is busy?** (8 points)

   Management thinks the kiosk helps more when the clinic is busy. Fit a model that lets the kiosk effect change with the number of patients waiting, say what you find, and decide whether to keep the extra term. Back that decision with a number.

7. **A number for one patient** (6 points)

   A patient is checking in now: 8 people ahead of them, 3 staff on shift, a moderate case, and the kiosk in use. Predict how long this visit will take, give the interval the front desk should quote, and say in one sentence why that interval rather than the other.

8. **What you would tell management** (4 points)

Write four to six sentences to the clinic manager. Say what you found, how sure you are, and what they can and cannot do with it. Assume they have not taken this course.